

Geological Section Profile — Complete User Guide

A step-by-step guide to all parameters, processing logic, outputs, and practical considerations.

1. Tool Overview

Geological Section Profile is an ArcGIS Pro Python Toolbox tool designed to extract a topographic profile from a DEM along a geological section line and create geological markers in an Elevation–Distance coordinate system.

The tool samples a DEM along a geological section line, builds a topographic profile in Elevation–Distance coordinates, and creates markers for intersections with lithological units, faults, and optional line or polygon features.

The tool does not automatically interpret subsurface geology. It extracts boundaries and intersections, but determining true layer thickness, geological dip and strike, fault displacement, subsurface geometry, and other interpretations remains the responsibility of the user.

2. General Parameter Order

1. Lithology Polygon Layer — Required
2. Lithology Name Field (Optional) — Optional
3. Other Polygon Feature (Optional) — Optional
4. Other Polygon Name Field (Optional) — Optional
5. Fault Line Layer — Required
6. Fault Name / ID Field (Optional) — Optional
7. Other Line Feature (Optional) — Optional
8. Other Line Name / ID Field (Optional) — Optional
9. Geological Section Line — Required
10. DEM Raster — Required
11. DEM Sampling Interval — Required
12. Section Layout Scale — Required
13. Marker Height (% of Elevation Range) — Required
14. Reverse Section Direction — Required
15. Create Lithology Boundary Markers — Required
16. Create Fault Intersection Markers — Required
17. Output Folder — Required
18. Output Base Name — Required

3. Detailed Parameter Descriptions

Lithology Polygon Layer — 1

Data type: GPFeatureLayer with Polygon geometry. Required.

The main polygon layer representing rock units, formations, members, deposits, or mapped units. The tool intersects this layer with the Geological Section Line and extracts locations where the section crosses lithological boundaries.

If Lithology Name Field is selected, the unit name is used for related Lithology outputs and labels.

The input must be Polygon geometry. Point or Polyline layers are not accepted. Logical topology and correct boundaries are recommended; gaps, overlaps, and very small polygons can create additional intersections.

Lithology Name Field — 2

Data type: Field. Optional. Filter: String.

Contains the name or code of the rock unit. When selected, the value is used for the NAME attribute in related Lithology outputs and, when possible, a Lithology label is placed at the actual midpoint of the unit along the section.

If left empty, Lithology boundaries can still be extracted, but named unit labels are not generated.

Examples: LITHOLOGY, UNIT_NAME, FORMATION, CODE.

Other Polygon Feature — 3

Optional Polygon layer whose intersection with the section should also be displayed.

Examples include alteration zones, mining areas, protected areas, geomorphological units, or other Polygon features. This layer does not replace Lithology; it is processed separately.

Other Polygon Name Field — 4

Optional name or ID Field for Other Polygon Feature. Select the Other Polygon Feature first so its fields become available.

Fault Line Layer — 5

Data type: GPFeatureLayer with Polyline geometry. Required.

Contains faults or other lines that should be represented as Faults in the section.

The tool finds each Fault intersection with the section, calculates its Chainage or Distance from the beginning of the section, and obtains the elevation at that position from the DEM.

If a Fault and Lithology Contact occur at the same Chainage, both markers are retained; the Lithology marker is positioned above the Fault marker.

Fault Name / ID Field — 6

Optional name or ID Field for the Fault. Used to store the Fault name or identifier in the output.

Examples: FAULT_ID, NAME, F_CODE.

Other Line Feature — 7

Optional Polyline layer for line features other than Faults.

Examples include rivers, roads, lineaments, specific geological contacts, structural boundaries, or other line features. Its outputs are stored separately from Fault outputs.

Other Line Name / ID Field — 8

Optional name or ID Field for Other Line Feature. If required, the selected value is stored in the output.

Geological Section Line — 9

Data type: GPFeatureLayer with Polyline geometry. Required.

This is the main geometry defining the actual geological section path. Its length defines the horizontal Distance axis. The DEM is sampled along this line, and all geological intersections are measured relative to it.

A suitable Projected Coordinate System is recommended for accurate distance calculations. If the Coordinate System is geographic, the tool warns that a Projected CRS is recommended.

The direction of this line defines the default section direction; Reverse Section Direction can be used to reverse it.

DEM Raster — 10

Data type: GPRasterLayer. Required.

The Digital Elevation Model provides ground-surface elevation along the section.

At each sampled position, the elevation is read from the DEM. The resulting Elevation–Distance points form the topographic profile.

The DEM must cover the entire section. Extensive NoData values or areas outside the DEM can reduce valid samples and may prevent profile creation.

DEM Sampling Interval — 11

Data type: GPDouble. Required. Default: 10.0.

Sampling distance along the section, normally interpreted in meters.

Smaller values create more samples and more detail but can increase processing time. Larger values increase speed but may miss small topographic features.

The tool also considers DEM cell size when controlling the effective sampling step.

Section Layout Scale — 12

Data type: GPDouble. Required. Default: 10000.0.

The denominator of the section drawing scale. 10000 means 1:10,000.

It is used to construct the Polygon below the topographic profile. Polygon depth equals $0.04 \times \text{scale denominator meters}$.

Examples:

- 1:5,000 → 200 m
- 1:10,000 → 400 m
- 1:25,000 → 1,000 m

This is a layout geometry and does not represent true geological thickness or true subsurface depth.

Marker Height (% of Elevation Range) — 13

Data type: GPDouble. Required. Default: 3.75.

Defines marker height as a percentage of the profile elevation range.

Marker Height = Elevation Range × Percentage ÷ 100

The parameter controls display size only; it is not true geological thickness.

Reverse Section Direction — 14

Data type: GPBoolean. Required. Default: False.

Reverses the processing direction of the section. When False, the original Geological Section Line direction is used. When True, Distance and unit order are reversed.

Create Lithology Boundary Markers — 15

Default: True.

Controls whether Lithology boundary intersections are created as markers. It affects marker creation only; DEM profile extraction is independent.

Create Fault Intersection Markers — 16

Default: True.

Controls whether Fault intersections are created as markers. If Fault and Lithology Boundary occur at the same Chainage and both options are enabled, both markers are retained.

Output Folder — 17

Required. Folder where all outputs are stored. It must exist and be writable.

Output Base Name — 18

Required. Default: Section.

Base name used for the output set. For example, outputs may be named Topo_Section, Section_TopoPoly, Section_Lith, Section_LithLabel, and Section_Fault.

The tool cleans and shortens output file names to improve compatibility with Shapefile naming limitations and the 10-character Field-name limitation.

4. Tool Processing Logic from Start to Finish

1. Input geometry types are validated: Lithology must be Polygon; Fault and Geological Section must be Polyline.
2. The Geological Section Line is read and reversed when requested.
3. Its Coordinate System is used as the spatial reference for output geometries.
4. The DEM is sampled along the section.
5. Samples are converted to Elevation–Distance coordinates.
6. A Topographic Profile is created.

7. A TopoPoly is constructed below the profile using Section Layout Scale.
8. Lithology, Fault, and optional feature intersections are identified.
9. Distance or Chainage is calculated for each intersection.
10. Elevation is obtained from the DEM at each intersection.
11. Markers are created in Elevation–Distance space: X = Distance and Y = Elevation.
12. When Lithology Name Field is selected, a label is created at the midpoint of the actual section interval inside each Lithology unit.
13. If Lithology Boundary and Fault share a Chainage, both are retained and the Lithology marker is placed above the Fault marker.
14. When available, `.lyrx` files for output styling and labeling are created.

5. Coordinate System and Elevation–Distance Space

Marker geometries are not stored in the original map coordinates. They are created in Elevation–Distance space for section drawing.

X represents distance from the beginning of the section. Y represents elevation.

Therefore, markers should not be expected to overlay the original map at their X and Y values. They are intended for geological section drawing.

For accurate distance calculations, a suitable Projected Coordinate System is recommended.

6. Lithology Label Logic

Lithology boundaries remain at their actual Contact positions. Unit-name labels are not placed directly on Contacts.

The tool finds the actual section interval inside each Polygon and calculates its midpoint.

Example: if a unit extends from Distance 120 to Distance 460, its label is placed at approximately Distance 290.

The label elevation is taken from the topographic profile at the same Distance.

7. Main Outputs

- Topo — PolyLine Shapefile — Topographic Elevation–Distance profile
- TopoPoly — Polygon Shapefile — Polygon below the profile for layout
- Lith — PolyLine Shapefile — Lithology boundary markers
- LithLabel — Point Shapefile — Lithology unit label points
- Fault — PolyLine Shapefile — Fault intersection markers
- OtherP — PolyLine Shapefile — Other Polygon markers when used
- OtherL — PolyLine Shapefile — Other Line markers when used
- Layer File (*.lyrx) — Style or label output when available

8. Important Output Fields

ID — Internal feature identifier

TYPE — Marker type, such as LITH_BOUND or FAULT

NAME — Source Feature name or ID, when available

M_DIST — Distance or Chainage from the beginning of the section

M_ELEV — Elevation at the marker location

MAP_X — X coordinate of the original intersection point

MAP_Y — Y coordinate of the original intersection point

SRC_OID — Source Object ID

ELEV_BOT — Lower elevation of the profile Polygon, when available

9. Recommended Starting Settings

Lithology Polygon Layer: rock-unit layer

Lithology Name Field: unit-name Field, such as LITHOLOGY

Fault Line Layer: fault layer

Geological Section Line: section line

DEM Raster: suitable DEM covering the area

DEM Sampling Interval: 10 m or a value consistent with DEM cell size

Section Layout Scale: 10000 for 1:10,000

Marker Height: 3.75%

Reverse Section Direction: False unless the direction should be reversed

Create Lithology Boundary Markers: True

Create Fault Intersection Markers: True

Output Base Name: Section

10. Numerical Example

Assume minimum elevation = 1250 m and maximum elevation = 1850 m. Elevation Range = 600 m.

With Marker Height = 3.75%:

$$600 \times 3.75 \div 100 = 22.5 \text{ m}$$

With Section Layout Scale = 10000:

$$\text{TopoPoly depth} = 10000 \times 0.04 = 400 \text{ m}$$

This 400 m is a layout depth and does not represent true geological thickness.

11. Common Errors and Solutions

- Incorrect geometry type: Lithology must be Polygon; Fault and Section must be Polyline.
- Fewer valid DEM samples: DEM must cover the section and the path should not contain excessive NoData.

- DEM Sampling Interval must be greater than 0.
- Section Layout Scale must be greater than 0.
- Marker Height percentage must be greater than 0.
- Output Folder must exist before execution.
- If a Field cannot be selected, select the related Layer first.
- Marker coordinates are in Elevation–Distance space; this is expected.

12. Important Tips for Output Quality

Use a Projected CRS for accurate distance measurements.

Consider DEM cell size and avoid unnecessarily small Sampling Intervals.

Check the section path for NoData before running the tool.

Select Lithology Name Field when unit names should be displayed.

Check Geological Section Line direction and Reverse Section Direction.

Marker Height controls only visual marker size and is not true geological thickness.

Use a separate Output Base Name for each section.

13. Real Data vs Section Display

The Topographic Profile is extracted from the DEM and represents the ground surface along the section.

Lithology and Fault markers show where surface Features intersect the Section and do not, by themselves, define subsurface geometry.

For a final geological section, field observations, boreholes, geological dip and strike, geological structures, and cross-cutting relationships must be considered during manual interpretation.

14. Pre-Execution Checklist

- Lithology Polygon Layer selected
- Lithology Name Field selected when needed
- Fault Line Layer selected
- Fault Name / ID Field selected when needed
- Geological Section Line is a valid Polyline
- DEM covers the entire section path
- DEM Sampling Interval is greater than zero
- Section Layout Scale matches the intended scale
- Marker Height is positive and reasonable
- Section direction checked

■ Output Folder exists and is writable

■ Output Base Name defined

Geological Section Profile — Complete Help Guide